

Lab 7 - Soft Sensing

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I. OBJECTIVES AND EXPERIMENTAL SETUP

The purpose of this lab is to be able to measure changes in the shape of soft materials. This was accomplished through the use of capacitive and resistive sensing of conductive fabric embedded within the silicon polymer. It was also done with the use of a pressure sensor.

To begin, we created molds for the initial thickness of the material. This involved a very thin layer of Dragon Skin 10 to act as a base for the two layers of fabric of the capacitive resistor. The conductive pieces of fabric were cut into rectangles with a little tab on one side. This fabric was then attached to the initial layer by spreading a thin film of polymer on the bottom and the top before letting it cure in the oven. This process was then repeated on the other side after curing with the other piece of conductive fabric. The purpose of the top layer of polymer was to insulate the fabric from the outside so that nothing changed its conductivity through contact. For the resistive sensor, the same process was used to create the initial thickness layer. Next, the resistive sensor was attached with the same method of a thin polymer film on the top and bottom before curing it in the oven.

For testing, we used multiple methods to see how each sensor reacted to different types of forces. For both sensors we put successively higher weights onto them to see how compression of the layers affected the output values. In addition to the compression test, we attached weights to the end of the sensor while keeping the opposing end attached to the table. This caused both sensors to stretch and altered the values of the output. This was done in 5 increments for each sensor.

For the pressure sensing portion of the experiment, we reused the pneu-net actuator from lab 3. This was connected to the pressure sensor and wired to the arduino to be able to measure the changes in pressure caused by pushing the actuator to different angles. The pneu-net actuator was moved to five different positions in order to see how the angles affect the amount of pressure in the actuator. These angles were 0, 30, 45, 60, 90 degrees.



Figure 1: Capacitive Sensor



Figure 2: Resistive Sensor

As a result of our testing in this experiment, we found that the capacitive resistor showed clear changes when it was affected by compressive forces while the resistive sensor was better equipped to measure changes in length from stretching. We also found that depending on the direction of bending, the pressure in the actuator would either decrease or increase. Because we compressed our actuator, the pressure increased with each increase in angle.

II. RESULTS & ANALYSIS

A. CAPACITIVE SENSING

The capacitive sensor ended up being relatively responsive to both tension and compression deformations. Figure 3 shows that the relationship between the capacitance and an applied force is relatively linear. The nonlinearity that is seen in figure 3c is probably due to the sensor plate not stretching uniformly during the tension tests. Qualitatively,

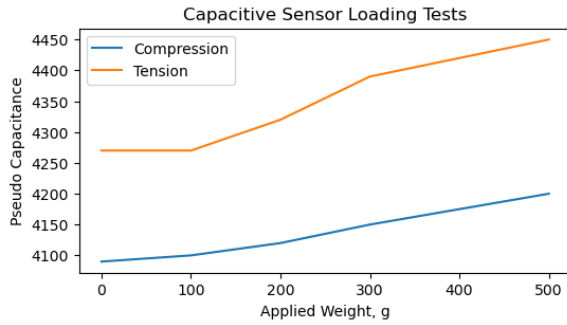


Figure 3a: Capacitance vs Applied Load

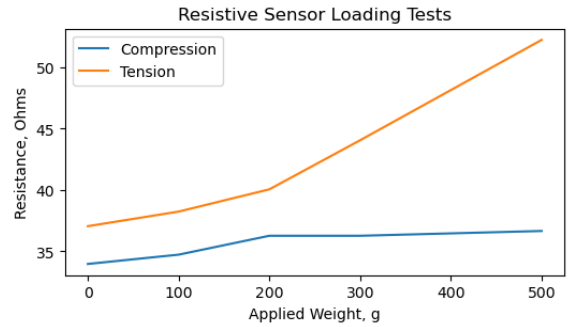


Figure 4a: Resistance vs Applied Load

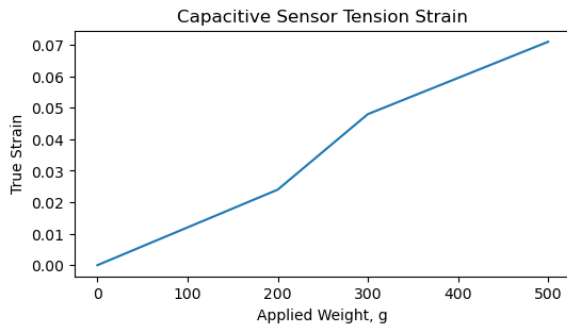


Figure 3b: Strain vs Applied Load

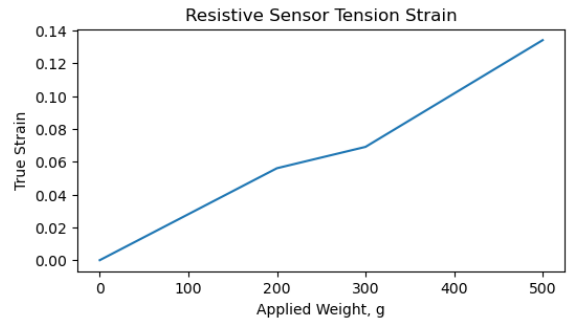


Figure 4b: Resistance vs Applied Load

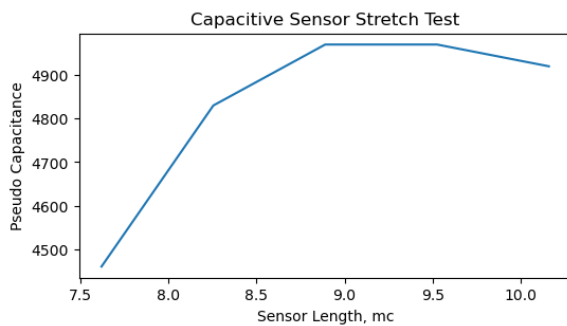


Figure 3c: Capacitance vs Stretched Length

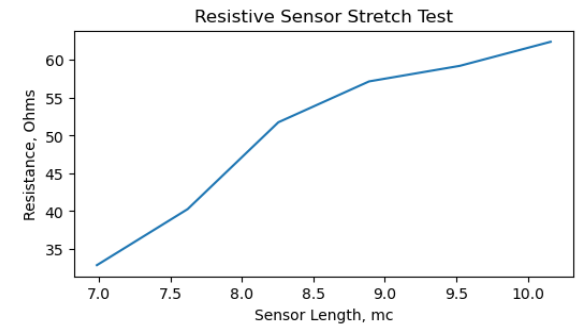


Figure 4c: Resistance vs Stretch Length

the capacitive sensor had a moderate sensitivity to shear deformation, which is probably due to the coupling with compression and tension deformation.

B. RESISTIVE SENSING

The resistive sensor was fairly responsive to tension deformation, but had a poor responsiveness to compression deformation. This difference was expected due to the resistive element deforming more when under tension than when under compression. Figure 4 shows that the relationship between the resistance and the deformation is pretty linear, even during the compressive deformation test.

The resistive sensor had a relatively poor response to shear deformation, which is probably due to the small amount of deformation that the resistive element experiences.

C. PRESSURE SENSING

The pressure sensor had a linear relationship, as seen in figure 5, between the bending angle and pressure measurement. The only exception to this is at large bending angles, greater than 60 degrees, where the sensor's shape becomes less of a smooth curve.

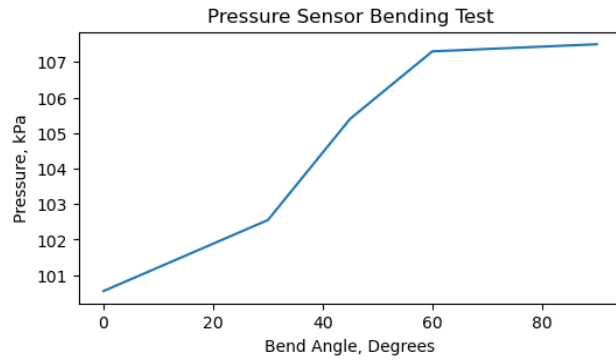


Figure 5: Pressure vs Bending Angle

D. ANALYSIS

The capacitive sensor's output contained some noise from both our physical presence and from the metal weights that we used to apply a load. One way to reduce this noise would be to build a shielding mechanism into the sensor, such as a grounding plate, that would absorb some of the noise. Another way to reduce the noise would be to apply filtering of some kind on the output signal. This could reduce small fluctuations that are caused by the added noise.

The resistive sensor generally had a larger relative change in value compared to the capacitive sensor. However, the lab demonstrated that the capacitive sensor was more effective for detecting compressive forces than the resistive sensor.

The geometries of both sensors could be fine-tuned for a desired application or behavior. The capacitive sensor could be modified by adding multiple layers of plates to amplify the compressive measurement or the distance between the plates could be modified to change the possible capacitance values. The resistive element can be shaped differently, such as a spiral, to enhance measurement in a specific direction or the element could be doubled up to amplify the resistance change for a given deformation.

Due to both sensor measuring changes in geometry, they both have coupling between different types and directions of deformation. One way to limit this would be by adding strain limiting materials that would focus the deformation in a specific direction. Another solution could be to add additional sensors that measure in different directions so that the collective output can be evaluated. A final possible mitigation is to use sensors where one type of deformation

is the primary contributor, such as the resistive sensor where tension has a much larger impact than compression.

III. CONCLUSION

The lab demonstrated that capacitive sensors are effective for detecting compressive forces, with capacitance increasing as the plate area grows or as the plates get closer together. While resistive sensors excel at measuring elongation, as resistance increases when the cross-sectional area decreases, due to stretching, leading to a larger drop in voltage. However, the resistance sensor is not ideal for measuring compression.

Pressure sensing with the pneu-net actuator revealed that internal pressure increases with bending angle, confirming expected pneumatic behavior. Overall, the findings highlight the importance of precise material preparation and standardized testing to improve reliability and consistency in soft material characterization for soft robotics applications.

Experimental Video:

<https://drive.google.com/file/d/1hx CikOHrFjTlabTX1lLe0fEM9WjHziPZ/view?usp=sharing>